

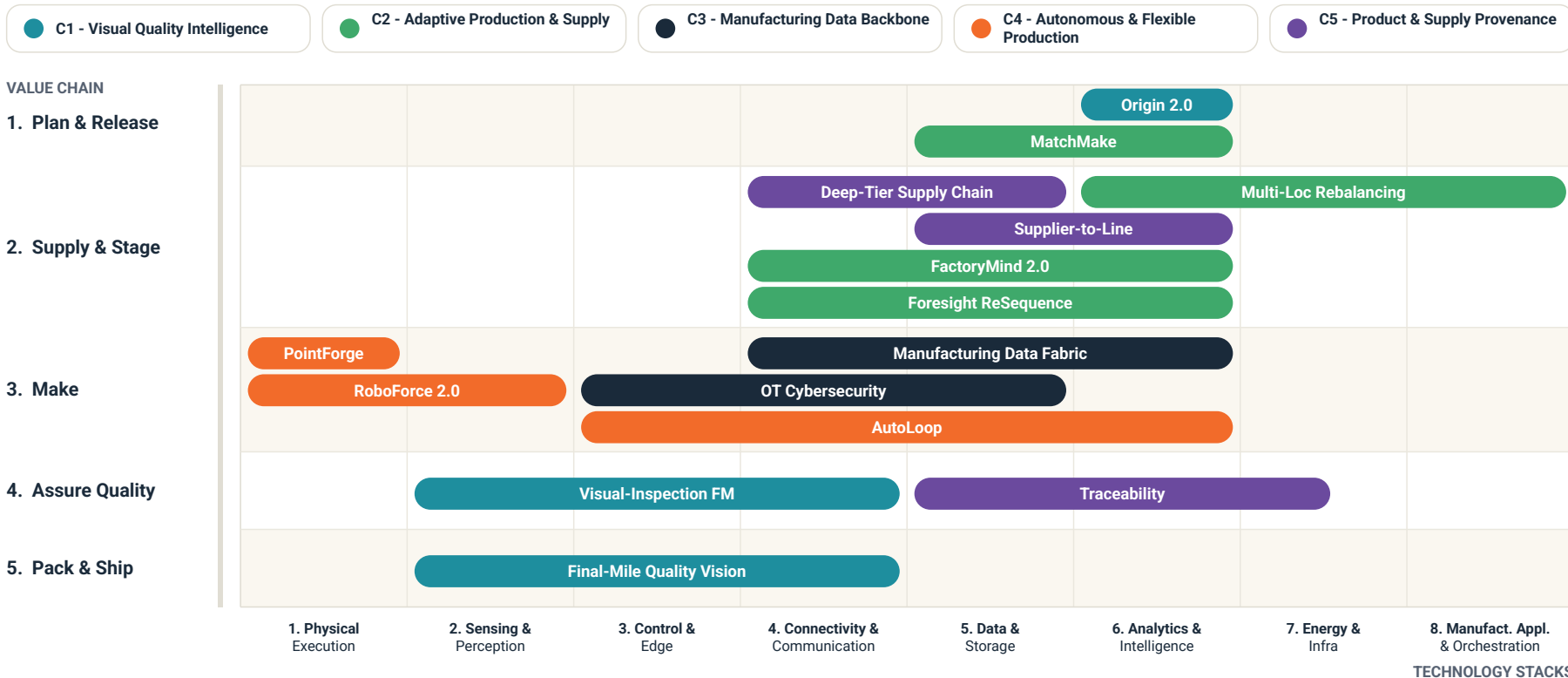
16 investable concepts · 5 strategic themes

These are the clusters & venture concepts to be matched with founders before the Hackathon, **please submit your preference via the [Google Form](#) by 11 Sep.**
To review the detailed information for the concept of your interest, [please access this folder](#). We'll walk through the concepts on the Kick-off call, for any questions.

01 Quality Intelligence 3 concepts	02 Adaptive Production & Supply 4 concepts	03 Manufacturing Data Backbone 3 concepts	04 Autonomous & Flexible Production 3 concepts	05 Product & Supply Provenance 3 concepts
01 Visual-Inspection FM A manufacturing-tuned vision foundation model.	04 Foresight ReSequence Supply-signal-driven floor re-sequencing in hours, not weeks.	08 Manufacturing Data Fabric Vendor-neutral unified data namespace across MES, PLC, ERP, LIMS.	11 AutoLoop AI writes setpoints back to the line inside a certified envelope.	14 Production to Field Traceability End-to-end per-unit product traceability platform.
02 Final-Mile Quality Vision AI-vision at dispatch. damage, label, integrity.	05 FactoryMind 2.0 Retiring planners' know-how, captured into an operational co-pilot.	09 OT Cybersecurity Managed detection & response purpose-built for connected manufacturing.	12 RoboForce 2.0 Outcome-priced robotic workforce. Pay per finished part.	15 Deep-Tier Supply Chain T2-T4 supplier verification for EUDR / CBAM / CSDDD.
03 Origin 2.0 Manufacturability intelligence at design time.	06 MatchMake Neutral marketplace matching free industrial capacity to parts demand.	10 Value Chain Data Weaver Reveal how operations run. Then optimize the value chain.	13 PointForge Container-based micro-factories at point of need.	16 Supplier-to-Line Data 2 sided platform for reusable suppliers, with ASN accuracy.
	07 Multi-Loc Inventory Cost-justified inter-site inventory rebalancing across a plant network.			

SDM Venture Portfolio - Landscape overview of concepts

SDM concepts for the Accelerator Program cover multiple technology layers & value chain stages, addressing near-term ROI and strategic positioning



C1 Quality Intelligence

3 ventures

AI-native quality intelligence across the full product lifecycle — catch issues at design, detect them on the line, defend them at dispatch.

VENTURE CONCEPTS

01

Visual-Inspection FM

A manufacturing-tuned vision foundation model, pretrained on 200+ Bosch plants of visual data. Fine-tunes for a new SKU in hours with <100 labelled samples. Handles novel defects zero-shot and no rule-tuning per changeover.

02

Final-Mile Quality Vision

An AI-vision station at the dispatch bay that inspects every unit in <2 s and captures photographic evidence timestamped to the serial. Instant defence dossier on any warranty or cargo-damage dispute or theft.

03

Origin 2.0

A manufacturability intelligence engine that gives engineers instant, DFM feedback grounded in real Bosch process data. Where conflicts arise, a generative AI layer suggests design alternatives on top of trusted expert knowledge. The result: fewer late-stage redesigns and designs that are industrializable from day one.

WHY THIS THEME WINS

3 regs /
18 mo

Regulatory forcing function

PLD (Dec 26) · EU Machinery Reg (Jan 27) · EUDAMED (May 26) — per-unit visual evidence becomes personal liability. Non-optional spend.

5–20 %

Biggest untouched margin lever

Cost of Quality sits at 5–20 % of revenue for manufacturers. AI-native quality is the only category that touches prevention, appraisal and failure costs at once.¹

+30 %

Tech inflection has arrived

Vision foundation models beat rule-based systems by 30%+ and adapt to new SKUs zero-shot. Category becomes buildable at faster pace. ^{2,3}

WHITE SPACES

01

AI-native + plant-floor trust combo

Incumbents (Cognex, Keyence, ISRA) own trust; AI-native players (Landing AI, Instrumental) own the tech. No competitor combines both.

02

Full-lifecycle quality platform

No player spans design → line → dispatch on a single AI-native architecture. Category-defining opportunity is open.

03

Manufacturing-tuned foundation models

Vision FMs today are generic. A plant-data-trained FM sold as a manufacturing platform is greenfield — first mover captures the moat.

04

Cost-of-Quality as a CFO conversation

Quality AI today is sold to QA and plant managers. Nobody positions it to the CFO as the biggest margin lever — new buying center reframes the category.

C2 Adaptive Production & Supply

4 ventures

Sense supply disruption early, decide with captured expert judgement, and execute — re-sequenced floors, rebalanced stock, matched capacity.

VENTURE CONCEPTS

04

Foresight ReSequence

Probabilistic ETA forecasting flags long-lead component risk weeks early, auto-generates ranked contingency schedules and pushes re-sequencing to existing APS, MES and AGV/AMR fleet managers — nothing replaced.

05

FactoryMind 2.0

Extends FactoryMind to procurement, planning and logistics — capturing the tacit judgement behind expediting calls, exceptions and workarounds into a queryable AI copilot. A read-layer across ERP, TMS and WMS.

06

MatchMake

An AI-driven marketplace matching buyers to the right production capacity and capability, powered by Bosch's DFM know-how. It replaces manual matching with instant recommendations, and can expand into logistics, inspection / certification to deliver the finished part.

07

Multi-Loc Inventory

A live rebalancing engine treating every plant, warehouse and hub as one shared pool — spotting idle stock against building shortages and dispatching the transfer: which pallet, which truck, which day.

WHY THIS THEME WINS

5 days

Response gap is structural

83% of supply chains can't respond to a disruption within 24 h — the average crisis response takes 5 days. Downtime cost: \$2.3m/hr. Tariffs & trade volatility are also forcing real time planning. ¹

\$20–60m

Untouched working-capital lever

Excess inventory hidden across multi-ERP sites while stockouts persist elsewhere; carrying cost runs 20–30% of stock value annually. ²

45% over 45

Tech meets demographic cliff

AI makes risk scoring, tacit-knowledge capture and capacity matching viable — just as veteran planners retire and take the playbook with them.

WHITE SPACES

01

Signal-to-floor middleware

Visibility tools stop at the plant gate; fleet tools see nothing upstream. No player closes procurement → scheduling → floor in one loop.

02

Transfer-level prescription

MEIO says what to hold, WMS moves inside four walls, TMS between them — nobody prescribes which pallet, which truck, which day, with owned execution.

03

Tacit-judgement capture

KM tools surface documents; procurement suites log transactions. Nobody captures the undocumented workaround expertise of supply chain teams.

04

Operator-owned marketplace liquidity

Every capacity marketplace brokers a third-party network; none seeds AI matching with its own plant footprint and production know-how.

C3 Manufacturing Data Backbone

3 ventures

The infrastructure category of modern manufacturing — the invisible foundation on which every intelligent, connected factory is built.

VENTURE CONCEPTS

08

Manufacturing Data Fabric

A vendor-neutral unified namespace layer on top of existing plant systems (MES, PLC, ERP, LIMS, historian). Standard schema across plants. Every SDM app runs on it. Every new deployment compounds the schema moat.

09

OT Cybersecurity

An MDR service purpose-built for connected manufacturing — plant-native sensors, OT-protocol-aware analytics, 24/7 SOC operated with Bosch cyber expertise. Compliance-ready for NIS2 and CRA. Deployable as a drop-in monitor without disrupting production.

10

Value Chain Data Weaver

An AI-powered discovery platform that automatically reconstructs real operational value chains across fragmented enterprise systems and processes — from first input to final output — without months of interviews, manual process mapping, or moving customer data into a third-party platform.

WHY THIS THEME WINS

**NIS2
CRA**

Regulatory & liability clock

NIS2 in force · CRA lands Sept 2026 · IEC 62443 continuously tightens — plant cyber and data governance now carry personal executive liability.

40–60 %

The silent tax on every AI project

Every manufacturing AI project spends 40–60 % of implementation time wrangling plant data. A unified infrastructure erases this tax at the industry level.

+300%

Attack surface has tripled

OT-targeted ransomware attacks up 300 %+ since 2021. Manufacturing is now Europe's most-attacked critical sector — a single event routinely costs €300 M+ in downtime. ^{5, 2}

WHITE SPACES

01

Vendor-neutral data substrate

MES (manufacturing execution system) vendors (Siemens, Rockwell, Nexeed) bundle their namespace; cloud players (AWS, Azure) push their own schemas. A genuinely neutral substrate is unclaimed.

02

OT-native cybersecurity

Current MDR (Managed Detection and Response) services are IT-native, retrofitted onto OT. Nobody has built OT MDR from the ground up with protocol-aware sensors and manufacturing workflows.

03

Data + cyber as one platform

Every existing player sells data OR cyber, never both. Manufacturers need one substrate — the data layer needs the security layer to actually work.

04

CISO as a first-class buyer

Cyber vendors sell to CISOs but treat plant floor as afterthought; MES vendors ignore CISOs. Nobody positions the substrate as a CISO-level conversation.

C4 Autonomous & Flexible Production

3 ventures

Production that reconfigures itself — AI closing the control loop, robots priced by output, and cells that redeploy in hours instead of months.

VENTURE CONCEPTS

11 AutoLoop

An AI control layer that reads live process data and writes optimised setpoints back to the line inside safety-certified guardrails. Priced on realised yield and energy gains — monitored manufacturing becomes self-optimising.

12 RoboForce 2.0

RoboForce 2.0 delivers on-demand robotic workforce as a service — deploying task-specific robots into critical process roles with pricing tied to output, not fixed leasing. The same fleet is continuously reprogrammed and re-tooled as tasks change, removing the CapEx barrier and integration delays of traditional automation.

13 PointForge

PointForge turns Bosch's proven production blueprints into deployable, container-based micro-factories that manufacture parts at the point of need — factory floor, disaster zone, or forward base. Instead of slow, vulnerable supply chains, the same platform reconfigures in software to serve both commercial ramp-ups and defense sustainment.

WHY THIS THEME WINS

Jan 2027

Regulatory clock for autonomy

EU Machinery Reg 2023/1230 applies 20 Jan 2027 — the first EU law with explicit AI safety-function and autonomous-machinery rules.

\$1.4tn

Downtime and yield gap

Unplanned downtime costs the Fortune Global 500 ~\$1.4tn/yr — 11% of revenue, up from 8% in 2020. A 1% yield drop runs ~\$500k per line per year. ²

\$55.8bn

Physical-AI capital wave

Robotics companies raised \$55.8bn in 2026, clustering around AI models, data and deployment infrastructure. ²

WHITE SPACES

01

Certified closed-loop write-back

Analytics monitor, copilots suggest, APC incumbents advise inside their own control layer. No player closes the loop with safety-rated write-back on open hardware.

02

Outcome-priced robot labour

RaaS providers lease generic hardware by the month regardless of output. Nobody combines outcome pricing, on-demand tooling and retrofit-or-new in one contract.

03

Zero-infrastructure cell connector

Fixed-track and planar-motor systems embed infrastructure into the table; AMRs still need a handling step at each stop. Nothing links mixed-vendor cells in hours.

04

Blueprint-backed deployable cell

Defense entrants build from a startup base without manufacturing-engineering depth or a digital-twin blueprint library to redeploy rather than rebuild.

C5 Product & Supply Provenance

3 ventures

One verified record of what a unit is made of and where it came from — joined at the serial, provable to a regulator, reusable across buyers.

VENTURE CONCEPTS

14

Production to Field Traceability

A federated traceability layer connects production data to each unit serial number, creating a reusable knowledge graph for warranty defence, recalls, DPP, EUDAMED, supplier chargebacks, and sustainability reporting. It lands where ROI is immediate, then expands as the serial-level dataset compounds.

15

Deep-Tier Supply Chain

A white-label, single-tenant platform each corporate deploys inside its own boundary — tracing and verifying T2–T4 suppliers against customs records, certification bodies, satellite imagery and on-ground audit. Shared verification standards, never shared customer data.

16

Supplier-to-Line Data

Suppliers publish one master profile — certifications, EDI setup, quality documents, delivery history — reusable across every customer they ship to. On top: ASN accuracy scored against real dock scans, and staging-risk prediction that warns about the suppliers and their credibility.

WHY THIS THEME WINS

Dec 2026

A regulatory wall, not a trend

EUDAMED (May 26) · PLD 2024/2853 (Dec 26) · EUDR (Dec 26) · Battery Passport (Feb 27) — CBAM already in its definitive phase.

\$58bn

Warranty is the funded entry point

Global auto warranty runs ~2.2% of sales — rivalling OEM R&D spend. Ford alone accrued \$6.29bn in 2024, up 33% year on year. ^{10 11}

6+

The evidence exists, joined by nothing

Machine, MES, quality, LIMS, ERP-lot and supplier records all carry the proof — but nothing joins them by unit serial number. ¹²

WHITE SPACES

01

Unit-serial spine, many applications

MES owns intra-plant genealogy, warranty tools own the claim workflow, compliance vendors go point-to-point per regulation. Nobody packages one serial-level layer underneath all of them.

02

Private-by-design deep-tier proof

Shared risk networks expose the supplier map to competitors; in-house diligence does not scale; S2P suites hold metadata, not primary-source evidence.

03

A supplier profile reusable across buyers

S2P suites own supplier master data, WMS (warehouse management system) owns receiving, MES owns the line. No one links reusable profiles, ASN reconciliation against real dock scans and staging risk.

04

Evidence as a shared asset

The same verified serial and supplier record answers warranty, recall, DPP and Scope 3 questions — today each team buys its own point tool and re-collects the data.